

Computer Networks (CS3001)

Sessional-I Exam

Course Instructor(s):

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Section(s): SE-(A,B,C,D,E,F,G)

Total Time (Hrs): 1

Total Marks: 60

Total Questions: 5

Date: Feb 25, 2025

Roll No

Course Section

Student Signature

Do not write below this line.

Attempt all the questions.

Attempt Question # 1 and 2 on Question paper provided space, remaining questions on answer sheet.

Question: 1	Line Coding	Marks: 10	Time: 10 minutes
CLO 4	Analyze working and performance of key technologies, algorithms and protocols		

Represent the Following data elements in signal elements using the following methods.

i. 00000011

	Manchester							

	Differential Manchester							

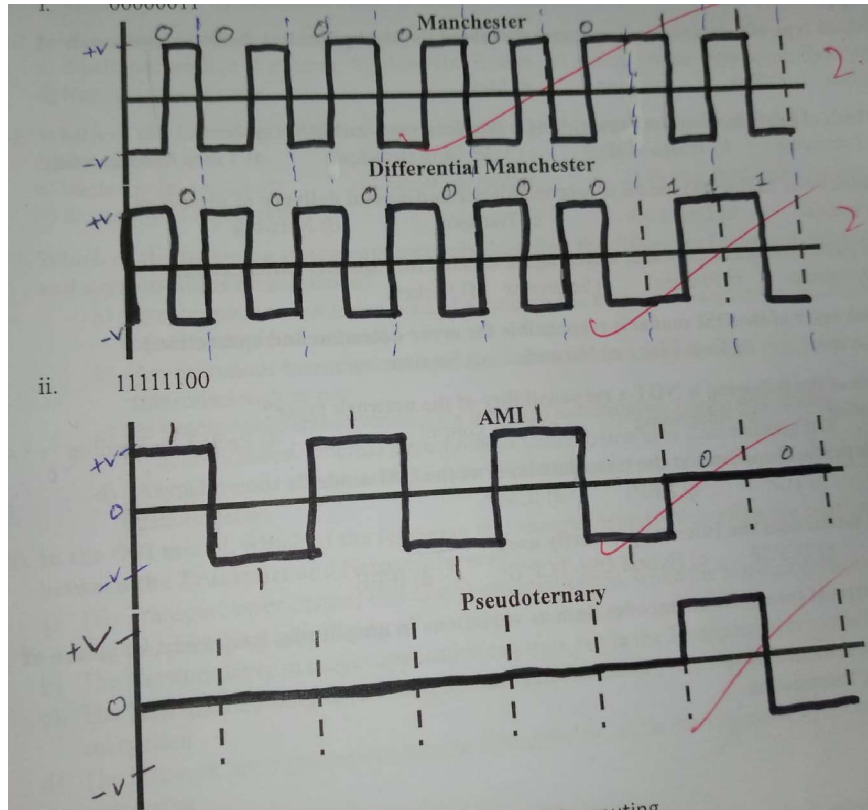
ii. 11111100

	AMI							

	Pseudoternary							

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Solution:



Question: 2	MCQs	Marks: 20	Time: 20 minutes
CLO 1	Describe the key terminologies and technologies of computer networks		

Sr. No.	Answer	Sr. No.	Answer	Sr. No.	Answer	Sr. No.	Answer
1	D	6	B	11	C	16	C
2	D	7	C	12	C	17	B
3	C	8	B	13	A	18	C
4	C	9	B	14	B	19	C
5	D	10	D	15	A	20	A

- Which of the following is NOT a fundamental characteristic of data communication?
a) Delivery b) Accuracy c) Timeliness d) Security
- Which type of network topology has a dedicated point-to-point link between every pair of devices?
a) Bus b) Star c) Ring d) Mesh

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3. **Which of the following is an example of a simplex communication system?**
a) Telephone b) Walkie-Talkie c) Television Broadcast d) Video Conferencing
4. **Which layer in the OSI model is responsible for end-to-end delivery of messages?**
a) Physical b) Data Link c) Transport d) Network
5. **In a packet-switched network, a message is divided into smaller units called:**
a) Datagrams b) Frames c) Segments d) Packets
6. **Which layer of the OSI model is responsible for error detection and correction?**
a) Transport b) Data Link c) Network d) Session
7. **Which of the following is NOT a responsibility of the network layer?**
a) Routing b) Addressing c) Flow control d) Packet forwarding
8. **Which protocol operates at the transport layer of the OSI model?**
a) IP b) TCP c) ARP d) ICMP
9. **Which model does the Internet primarily use?**
a) OSI b) TCP/IP c) Hybrid OSI-TCP/IP d) IEEE
10. **Which type of transmission encodes data as variations in amplitude, frequency, or phase of a signal?**
a) Baseband transmission b) Broadband transmission c) Digital transmission
d) Analog transmission
11. **In analog-to-digital conversion, which step involves sampling the analog signal at regular intervals?**
a) Quantization b) Modulation c) Sampling d) Encoding
12. **Which of the following is NOT a digital-to-digital encoding technique?**
a) Non-return-to-zero (NRZ) b) Manchester encoding c) Amplitude Shift Keying (ASK)
d) Differential Manchester
13. **What is the minimum bandwidth required to transmit a signal with a maximum frequency of 5 MHz?**
a) 5 MHz b) 10 MHz c) 2.5 MHz d) 1 MHz
14. **Which technique is used to detect and correct errors in digital transmission?**
a) Frequency Division Multiplexing b) Parity Checking c) Bandwidth Expansion
d) Modulation
15. **Which encoding scheme uses two levels of voltage for each bit without a return to zero?**
a) NRZ b) Manchester c) Differential Manchester d) Bipolar AMI
16. **Which of the following is an example of synchronous transmission?**
a) Morse Code b) Asynchronous Serial Communication c) Ethernet d) Radio Broadcast
17. **In a synchronous data transmission system, data is sent in:**
a) Small independent characters b) Fixed-size frames c) A single burst without timing
d) Random sequences

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18. Which of the following is NOT an advantage of digital transmission over analog transmission?
- a) Higher noise immunity b) Easier encryption c) Higher bandwidth requirements
d) Better signal regeneration
19. Which of the following statements correctly describes the difference between synchronous and asynchronous transmission?
- a) Synchronous transmission sends data one bit at a time, while asynchronous transmission sends multiple bits at once
b) Asynchronous transmission requires a dedicated clock signal, while synchronous transmission does not
c) In synchronous transmission, data is sent in a continuous stream with timing synchronization, whereas asynchronous transmission uses start and stop bits
d) Asynchronous transmission is more efficient than synchronous transmission for large data transfers
20. In the OSI model, which of the following statements correctly describes the relationship between the Transport and Network layers?
- a) The Transport layer ensures end-to-end communication, while the Network layer handles logical addressing and routing
b) The Network layer manages application requests, while the Transport layer routes data packets
c) The Transport layer is responsible for error detection, while the Network layer provides encryption
d) The Network layer guarantees reliable data delivery, while the Transport layer only focuses on addressing

Solve Following Questions on Answer Book

Question: 3	Latency	Marks: 15	Time: 15 minutes
CLO 1	Describe the key terminologies and technologies of computer networks		

A frame of size 5 million (5000,000) bits is being sent on a link with 10 routers each having a queuing time of 2 μ s and a processing time of 1 μ s. The length of the link is 2000 Km. The speed of light inside the link is 2×10^8 m/s. The link has a bandwidth of 5 Mbps.

- a) Find the total delay (latency) for the above communication. (10 marks)
b) Which component of the total delay is dominant? (2.5 Marks)
c) Which component is negligible? (2.5 Marks)

Solution:

Total Delay = Transmission Delay + Propagation Delay + Queuing Delay + Processing Delay

For Transmission Delay (T_{trans})

The transmission delay is calculated as:

Given:

- Frame Size = 5,000,000 bits
- Bandwidth = 5 Mbps = 5×10^6 bps

$$T_{trans} = 5,000,000 / 5 \times 10^6 = 1 \text{ second}$$

2. Propagation Delay (T_{prop})

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The propagation delay is calculated as:

Given:

- Distance = 2000 km = 2000×10^3 meters
- Speed of Light = 2×10^8 m/s

$$T_{\text{prop}} = 2000 \times 10^3 / 2 \times 10^8 = 0.01 \text{ seconds} = 10 \text{ ms}$$

3. Queuing Delay (T_{queue})

The queuing delay is the time the frame spends waiting in the queue at each router. Since there are 10 routers, the total queuing delay is:

$$T_{\text{queue}} = \text{Queuing Time per Router} \times \text{Number of Routers}$$

Given:

- Queuing Time per Router = $2 \mu\text{s} = 2 \times 10^{-6}$ seconds
- Number of Routers = 10

$$T_{\text{queue}} = 2 \times 10^{-6} \times 10 = 20 \times 10^{-6} \text{ seconds} = 20 \mu\text{s}$$

4. Processing Delay (T_{proc})

The processing delay is the time it takes for each router to process the frame. Since there are 10 routers, the total processing delay is:

$$T_{\text{proc}} = \text{Processing Time per Router} \times \text{Number of Routers}$$

Given:

- Processing Time per Router = $1 \mu\text{s} = 1 \times 10^{-6}$ seconds
- Number of Routers = 10

$$T_{\text{proc}} = 1 \times 10^{-6} \times 10 = 10 \times 10^{-6} \text{ seconds} = 10 \mu\text{s}$$

5. Total Delay (T_{total})

The total delay is the sum of all the delays:

$$T_{\text{total}} = T_{\text{trans}} + T_{\text{prop}} + T_{\text{queue}} + T_{\text{proc}}$$

Substituting the values:

$$T_{\text{total}} = 1 \text{ second} + 0.01 \text{ seconds} + 20 \times 10^{-6} \text{ seconds} + 10 \times 10^{-6}$$

$$T_{\text{total}} = 1.01003 \text{ seconds}$$

Dominant and Negligible Components

- Dominant Component:** The transmission delay ($T_{\text{trans}} = 1 \text{ second}$) is the dominant component because it is significantly larger than the other delays.
- Negligible Component:** The queuing delay (T_{queue}) and processing delay ($T_{\text{proc}} = 10 \mu\text{s}$) are negligible compared to the transmission and propagation delays.

Question: 4	impairment	Marks: 10	Time: 10 minutes
CLO 4	Analyze working and performance of key technologies, algorithms and protocols		

A railway communication system transmits signals over a 150 km network, but the signal degrades due to -0.4 dB/km attenuation, and an SNR of 5 dB. To ensure reliable data transmission,

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- a) How much amplification is needed
- b) what should be the minimum number of signal levels required for error-free communication?

Solution:

$$\begin{aligned} \text{Total Attenuation} &= \text{Attenuation per km} \times \text{Distance} \\ \text{Total Attenuation} &= -0.4 \times 150 \\ \text{Total Attenuation} &= -60 \text{ dB} \end{aligned}$$

- a) Amplification needed = +60dB
Shannon Theorem

$$C = B \log_2(1 + SNR)$$

Nyquist Theorem

$$\begin{aligned} C &= 2B \log_2 M \\ B \log_2(1 + SNR) &= 2B \log_2 M \\ \log_2(1 + SNR) &= 2 \log_2 M \\ 1 + SNR &= M^2 \end{aligned}$$

Effective SNR

$$\begin{aligned} SNR_{db} &= 10 \log_{10} SNR \\ SNR &= 10^{5/10} = 3.16 \\ M &= \sqrt{1 + SNR} \\ M &= \sqrt{1 + 3.16} \\ M &= 2 \text{ approx} \end{aligned}$$

Question: 5	OSI Layers	Marks: 5	Time: 5 minutes
CLO 1	Describe the key terminologies and technologies of computer networks		

Provide the name of the most relevant OSI layer (e.g. Physical Layer) with each of the following:

Sr. No.	Term	Layer
A	192.168.1.30	Network Layer
B	MAC address	Data Link Layer
C	SMTP	Application Layer
D	Session Management	Session Layer
E	Ethernet Cable	Physical Layer
F	Router	Network Layer
G	Chrome Browser	Application Layer
H	Encryption	Presentation Layer
I	Fiber Optics	Physical Layer
J	Port number	Transport Layer